

AI-Generated Image Detection for Secure Liveness Verification in Pathao Pay

Project Objective

This project aims to design and evaluate an AI-based detection system to distinguish real human-captured facial images from AI-generated or machine-manipulated images. The solution enhances Pathao Pay's selfie-based liveness verification, preventing identity fraud through synthetic or deepfake images.

Team Members

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Organization and External Mentor

Organization: Pathao Pay, Pathao Ltd. **Domain:** FinTech, Digital Identity Security

Domain Overview and Problem Statement

Advancements in generative AI and deepfake technologies pose significant risks to selfie-based identity verification systems. Attackers may bypass liveness checks using AI-generated facial images, leading to fraudulent account creation and security breaches. This project addresses automated detection of such synthetic images under real-world conditions.

High-Level Overview of Scope and Objectives

The project aims to:

- Study visual and statistical artifacts unique to AI-generated images
- Build a robust detection model resilient to real-world transformations
- Evaluate model performance against diverse fake and real image datasets
- Enhance the existing liveness verification system with additional security constraints
- Develop a prototype integration with Pathao Pay's verification pipeline

The scope includes model training, evaluation, system enhancement, and integration, but excludes full production deployment.

Justification of Project Complexity

This project is complex due to:

- Rapidly evolving generative models that continuously reduce detectable artifacts
- Conflicting requirements between high detection accuracy and low false positives
- Deep expertise required in computer vision, deep learning, and image forensics
- Robustness requirements against compression, resizing, and adversarial manipulation
- Real-time performance constraints in a financial verification system

Exploration of Solution and Implementation Methods

The solution involves training lightweight deep learning models, particularly MobileNet-based CNNs optimized for mobile deployment, alongside transformer architectures for complex analysis. A multi-layered verification approach will be implemented: primary detection runs on-device for real-time mobile verification, flagged cases escalate to secondary server-side analysis, and edge cases may require tertiary human review. Feature analysis will include frequency-domain artifacts, texture inconsistencies, and latent-space representations. The goal is to maximize automated mobile-first detection while minimizing human intervention, with the system exposed as an API integrated into Pathao Pay's workflow.